



Learning Sound Correspondences What About Neural Networks?

(aka a bumpy first year)

Clémentine Fourier



INRIA (ALMA_{na}CH)

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Studying the sound evolution of languages I

- Neogrammarian hypothesis: **Sound changes regularity**¹
If, at a given “point” in time, a phone in a given word changes into another phone, then all occurrences of the same phone in the same context also change in the same way.

Ex: [ka] Vulgar Latin > ... > [ʃa] French

- *cattus* *[kat.tʊs] > *chat* [ʃa] ‘cat’
- *blanca* *[blan.ka] > *blanche* [blɑ̃ʃ] ‘white (fem.)’
- *vacca* *[vak.ka] > *vache* [vaʃ] ‘white (fem.)’

Studying the sound evolution of languages II

- Special words: *cognates*

Given two languages with a common ancestor, two words are cognates if they are an evolution of the same word from said ancestor, called their proto-form.

- Comparative method

	A		B			C		D			E			F				
Sanskrit	y	u	g	a	m	dh	u	h	i	(tar)	s	n	u	ṣ (ā)	-	r	u	dh (iras)
Greek	z	u	g	o	n	th	u	g	a	(ter-)	-	n	u	- (os)	e	r	u	th (rós)
Latin	i	u	g	u	m	Ø	Ø	Ø	Ø	(Ø)	-	n	u	r (us)	-	r	u	b (er)
Gothic	j	u	k	-	-	d	au	h	-	(tar)	Ø	Ø	Ø	Ø (Ø)	Ø	Ø	Ø	Ø (Ø)
Gloss	'yoke'					'daughter'					'daughter-in-law'					'red'		

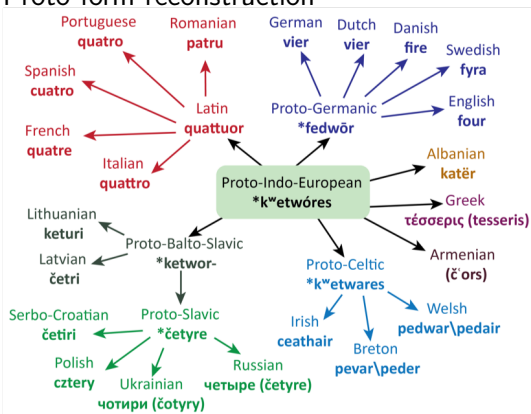
Figure 1: Regular sound correspondences across four Indo-European languages, illustrated with help of alignments along the lines of Anttila (1972: 246). In contrast to the original illustration, lost sounds are displayed with help of the dash “-” as a gap symbol, while missing words (where no reflex in Gothic or Latin could be found) are represented by the “Ø” symbol.

¹Hermann Osthoff and Karl Brugmann, *Morphologische Untersuchungen auf dem Gebiete der indogermanischen Sprachen*, volume 3 (S. Hirzel, 1880).



What are the "tasks" of phonological historical linguistics?

- Cognate and sound correspondences identification
- Proto-form reconstruction





How are cognate and sound change correspondences solved so far?

Automatic word comparison²

- Sound alignment + distance score (Levensthein, Turchin, ...)
- Sound alignment + clustering

²Johann-Mattis List, Simon J Greenhill and Russell D Gray, 'The potential of automatic word comparison for historical linguistics', *PloS one* 12:1 (2017).

³Philipp Koehn et al., *Moses: Open source toolkit for statistical machine translation*, in: *Proceedings of the 45th annual meeting of the association for computational linguistics companion volume proceedings of the demo and poster sessions* (2007).



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More recently, translation inspired methods

- Sound change prediction looks a lot like translation!
- Statistical translation methods (MOSES³)

²List et al. (as in n. 2).

³Koehn et al. (as in n. 3).

⁴Rama (as in n. 4).

⁵Dekker (as in n. 5).



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Where are the neural networks?

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Where are the neural networks?

- Siamese CNN⁴
- Perceptrons and Encoder decoders⁵ : '[The] structured perceptron model outperforms the encoder-decoder RNN.' ☹

²List et al. (as in n. 2).

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Goal

- Can we use neural networks to learn sound correspondences?
- On what kind of data?
- *Can we use reconstruct ancestral protoforms using neural networks?*

Looks like translation but small datasets, very small (and small vocabulary)

Why would we do that?

- Science! Understanding! (Much debate, much linguistics)
- Field linguistics⁶



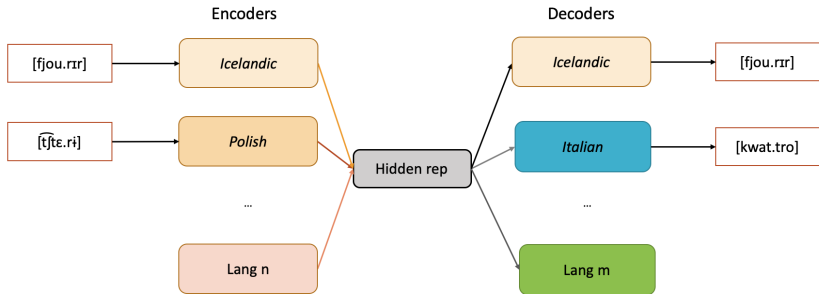
- Low resource translation with transfer?⁷

⁶Timotheus A Bodt and Johann-Mattis List, 'Testing the predictive strength of the comparative method: an ongoing experiment on unattested words in Western Kho-Bwa languages', *Papers in Historical Phonology* 4 (2019).

⁷Arbi Haza Nasution, Yohei Murakami and Toru Ishida, 'A Generalized Constraint Approach to Bilingual Dictionary Induction for Low-Resource Language Families', *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 17:2 (2017).



MEDeA - Multiway Encoder Decoder Architecture I

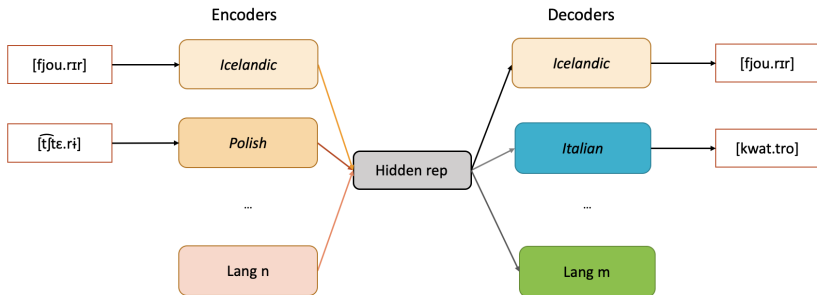


Possible model configurations

- Unidirectional/Bidirectional, GRU/LSTM
- Decoder with attention (Luong⁸ and Bahdanau⁹)



MEDeA - Multiway Encoder Decoder Architecture II



Losses - Accuracies

- Seq2seq (NLLLoss) - BLEU
- Siamese (Cosine Similarity) - Siamese (Cosine Embedding)

Implemented but not finished

- Transformers (will probably drop)
- Variational Encoders and Loss



Ease of use and reproducibility I

- Test suite and continuous integration
- Launched with JSON parameter files

```
# MODEL PARAMETERS
params_json = {
    "model_type": "CompositeModel",
    "submodels":
        [{
            "model_name": "Encoder", "lang": lang,
            "model_type": "GRU", "is_bidirectional": False,
            "num_layers": 1, "layer_dropout": layer_dropout,
            "uses_embedding_layer": True, "input_vocabulary_size": vocab_dim,
            "embedding_vector_size": emb_dim, "hidden_dimension": hidden_dim,
            "is_positive": True, "has_dense_output_layer": False
        } for lang in all_langs] +
        [{
            "model_name": "Decoder", "lang": lang,
            "model_type": "GRU", "is_bidirectional": False,
            "num_layers": 1, "layer_dropout": 0,
            "uses_embedding_layer": True, "input_vocabulary_size": vocab_dim,
            "embedding_vector_size": emb_dim, "hidden_dimension": hidden_dim,
            "output_dimension": vocab_dim, "has_dense_output_layer": True,
            "attention_type": AttentionType.LUONG_DOT
        } for lang in all_langs]
}
```



Ease of use and reproducibility II

- Everything is recorded...
 - Parameter files
 - Data shuffling
 - Model, embeddings, accuracies, loss, attention, precision, recall, ...
- Through several loggers
 - Console
 - File
 - Tensorboard
- Randomness is managed (all seeds fixed)
- Possibility for parallelism



Ease of use and reproducibility III

- Integration with Tensorboard

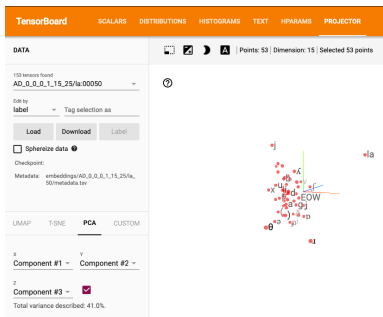


Figure 1: Embeddings

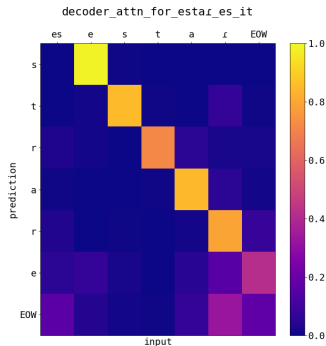
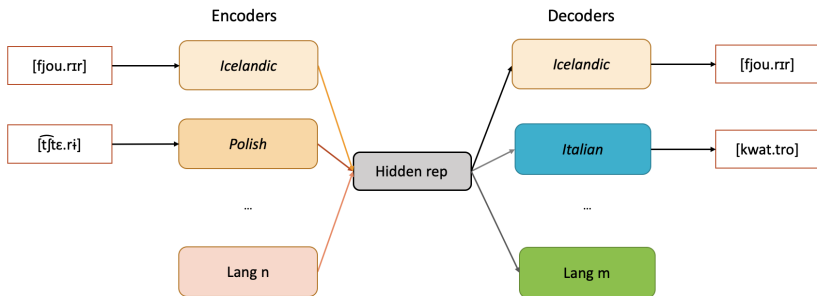


Figure 2: Attention

Reminder: This is our task



- Cognate "translation" to learn sound changes
- Baseline: MOSES
- Accuracy: BLEU



Preliminary experiment on Balto-Slavic Languages

- Data: cognates extracted from an etymological database

Dataset	PL-CZ	PL-LT	PL-IT
Cognate set	370	47	57

- Data sets so small that models cannot learn
- What kind of data carries the evolution of languages?



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Bilingual lexicon (train)	13,216	6,290	17,158

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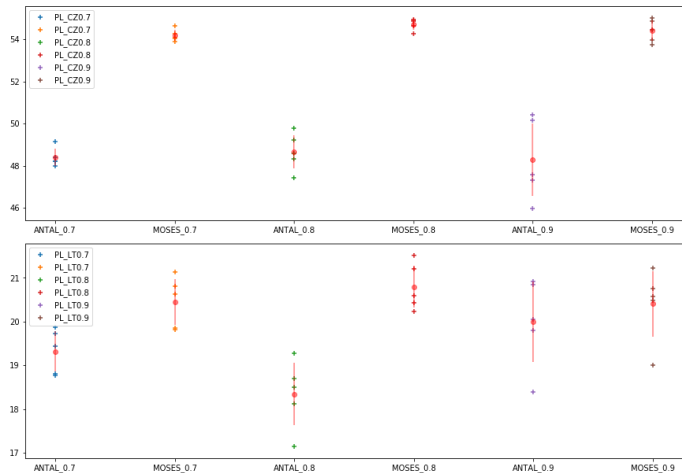


Figure 3: BLEU accuracy for PL-CZ (top) and PL-LT (bottom) cognates, for different train/validation splits and different shuffling seeds



Let's hunt problems!

Where do the learning problems come from for our NN?

- Cognate datasets too small for neural networks?
- Bilingual lexicons too noisy for our task?
- Neural networks not adapted for this task? 🤖



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What if we worked on artificial data?

- Can control size
- Can control noise level



Generating artificial data

- Creating a proto-language
 - Phone inventory (Romance languages)
 - Phonotactics (Classical Latin¹⁰)
- Creating daughter languages
 - Succession of phone changes (Romance: apocope, epenthesis, palatalisation, lenition, vowel prosthesis and diphthongisation)

Results: 20,000 word triplets

- [ʒolpast] > [ʒolbes], [ʒolpes]
- [ʃplutoi] > [isbledoi], [espletoi]

¹⁰András Cser, 'Aspects of the phonology and morphology of Classical Latin', Ph.D thesis, Pázmány Péter Katolikus Egyetem (2016).



Experiment

Task:

- Predict sequence of phones "translation", from PL to DL, DL to DL, DL to PL

Questions:

- Can MEDeA learn on a perfect (non-noisy) dataset?
- How does MOSES compare to MEDeA?
- What is the influence of the data size? (500, 1000, 1500, 2000, 3000)
- How do the models manage ambiguity? (1, 2, 3 best - corpus BLEU)
[Zolbes] < [Zolpast]



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[Zolbes] < [Zolpast] or [Zolpest] or [Zolpas] or [Zolpes] or ...

Comparing MOSES and MEDeA on artificial data I

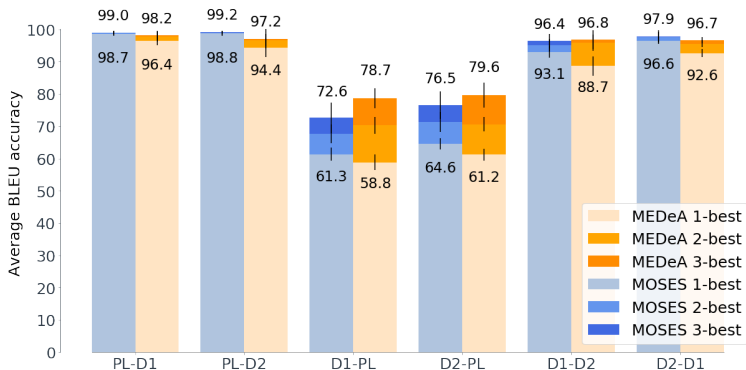


Figure 4: Comparing MOSES and MEDeA on different prediction directions

Comparing MOSES and MEDeA on artificial data II

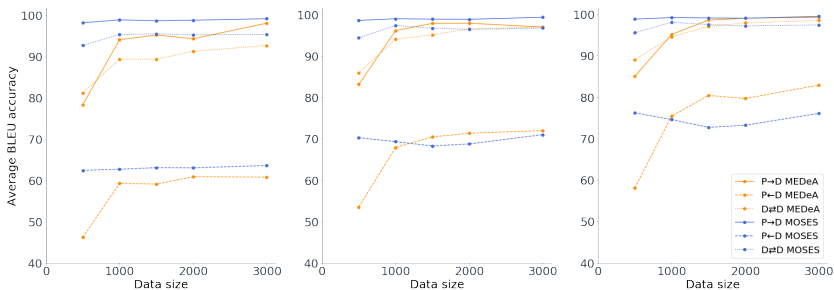


Figure 5: Influence of the data size and number of answers on the accuracy

We Need Cognates!

Database	Date	Relations	Relation types	Creation
EtymWordNet ¹¹	2013	550,000	Cognacy vs 'Other'	Wiktionary
CogNet ¹²	2019	3,000,000	Cognacy eq. to Bor!	Wiktionary Wordnet
EtymDB 1.0 ¹³	2017	500,000	Cognacy vs Bor vs Inh vs Comp vs Der	Wiktionary
COBL ¹⁴	2020 ?	?? ☹	Cognacy	Expert

¹¹Gerard de Melo, *Etymological Wordnet: Tracing The History of Words*, in: *Proceedings of the 9th Language Resources and Evaluation Conference (LREC 2014)* (2014).

¹²Khuyagbaatar Batsuren, Gabor Bella and Fausto Giunchiglia, *CogNet: A Large-Scale Cognate Database*, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (2019).

¹³Benoît Sagot, *Extracting an Etymological Database from Wiktionary*, in: *Electronic Lexicography in the 21st century (eLex 2017)* (2017).

¹⁴Cormac Anderson, Paul Heggarty and The CoBL Consortium, *Cognacy in Basic Lexicon database* (to be published).



Updating an Etymological Database: EtymDB 2.0

Method

- 2019 Wiktionary data dump > simplified xml
- simplified xml > csv data base
 - Word id - triplets: lexeme, gloss, language
 - Relation id - Relation type, word ids
- Problems with duplicate management:
Ex: παροιμίᾱ and παροιμία 'proverb'
=> grouping independently of diacritics

¹⁵Clémentine Fourrier and Benoît Sagot, *Methodological Aspects of Developing and Managing an Etymological Lexical Resource: Introducing EtymDB-2.0*, in: *Twelfth International Conference on Language Resources and Evaluation (LREC 2020)* (Marseilles, France, 2020).



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Result: Methodological paper and DB¹⁵

- 50% more lexemes and 40% more relations than EtymDB-1.0
- But English constitutes 50% of the DB
- 60% of the base has a gloss (meaning)

¹⁵ibid.



Experiment

Task:

- Predict sequence of phones "translation" for Romance languages - IT, ES, LA
 - LA-ES-IT: 605 cognate triples (1/3 for training, 2/3 for test)
 - LA-IT: 388 cognate pairs for train (extraction pb)
 - LA-ES: 296 cognate pairs for train (extraction pb)
 - IT-ES: 1764 cognate pairs for train
 - Monolingual data: 1500 words for train

Questions:

- Can MEDeA learn on a real (and noisy) dataset?
- How does MOSES compare to MEDeA?

Comparing MOSES and MEDeA on real data I

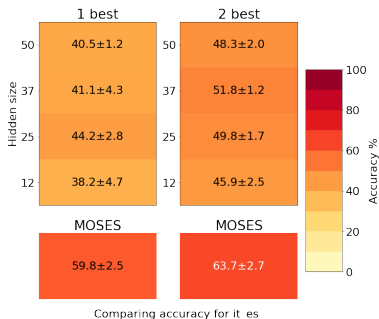


Figure 6: IT => ES

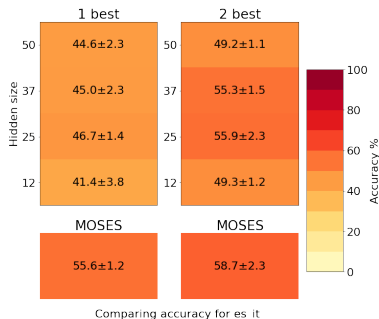


Figure 7: ES => IT

Data extraction problem: Some Spanish words appeared several times with different associations



Comparing MOSES and MEDeA on real data II

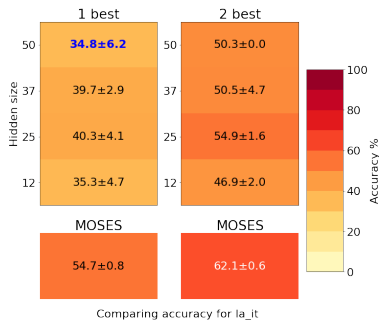
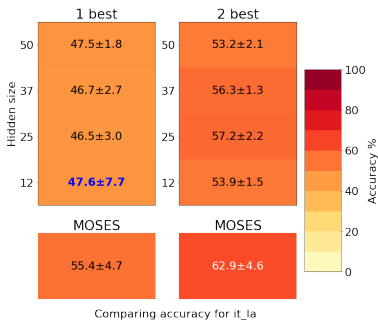


Figure 8: IT => LA

Figure 9: LA => IT

Problem with Latin: accusative or nominative



Current Status

- On ideal data
 - Neural networks perform better on ambiguity cases (DL=>PL)
 - Statistical models perform better on small datasets
- On real and very noisy data
 - Data size matters, data cleanliness too, and... ?
More data extraction and cleaning need to be done
(Spanish duplicates, Latin to DL sets, Latin ambiguity)
Monolingual data needs to be used to train MOSES's LM



And afterwards?

- What more can we learn from artificial data?
 - Use noisy artificial data
- What is the hidden representation hiding?
 - Shared attentions across all decoders
 - Blackbox study
- Can we develop new architectures capturing sound changes better?
 - Reversible networks
 - Deeper architectures

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References I

- Anderson, Cormac, Heggarty, Paul and The CoBL Consortium, *Cognacy in Basic Lexicon database* (to be published).
- Bahdanau, Dzmitry, Cho, Kyunghyun and Bengio, Yoshua, 'Neural machine translation by jointly learning to align and translate', *arXiv preprint arXiv:1409.0473* (2014).
- Batsuren, Khuyagbaatar, Bella, Gabor and Giunchiglia, Fausto, *CogNet: A Large-Scale Cognate Database*, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (2019), pp. 3136–3145.
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References II

- Cser, András, 'Aspects of the phonology and morphology of Classical Latin', Ph. D thesis, Pázmány Péter Katolikus Egyetem (2016).
- Dekker, Peter, 'Reconstructing language ancestry by performing word prediction with neural networks', *Master. Amsterdam: University of Amsterdam* (2018).
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References III

- Koehn, Philipp et al., *Moses: Open source toolkit for statistical machine translation*, in: *Proceedings of the 45th annual meeting of the association for computational linguistics companion volume proceedings of the demo and poster sessions* (2007), pp. 177–180.
- List, Johann-Mattis, Greenhill, Simon J and Gray, Russell D, 'The potential of automatic word comparison for historical linguistics', *PloS one* 12:1 (2017).
- Luong, Minh-Thang, Pham, Hieu and Manning, Christopher D, *Effective Approaches to Attention-based Neural Machine Translation*, in: *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing* (2015), pp. 1412–1421.

References IV

- Melo, Gerard de, *Etymological Wordnet: Tracing The History of Words*, in: *Proceedings of the 9th Language Resources and Evaluation Conference (LREC 2014)* (2014), pp.1148–1154.
- Nasution, Arbi Haza, Murakami, Yohei and Ishida, Toru, 'A Generalized Constraint Approach to Bilingual Dictionary Induction for Low-Resource Language Families', *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 17:2 (2017), pp. 9:1–9:29.
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References V

- Rama, Taraka, *Siamese convolutional networks for cognate identification*, in: *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers* (2016), pp. 1018–1027.
- Sagot, Benoît, *Extracting an Etymological Database from Wiktionary*, in: *Electronic Lexicography in the 21st century (eLex 2017)* (2017), pp. 716–728.